

```
package ques6LibraryManagementSystem;

public class Book {

    int bookId;
    String title;
    String author;

    public Book(int bookId, String title, String author) {
        this.bookId = bookId;
        this.title = title;
        this.author = author;
    }

    @Override
    public String toString() {
        return "Book ID: " + bookId +
            ", Title: " + title +
            ", Author: " + author;
    }

}
```

```
package ques6LibraryManagementSystem;
import java.util.Arrays;
import java.util.Comparator;

public class LibraryManager {

    // Linear Search
    public static Book linearSearch(Book[] books, String title) {

        for (Book book : books) {

            if (book.title.equalsIgnoreCase(title)) {
                return book;
            }
        }

        return null;
    }

    // Sort Books by Title
    public static void sortBooks(Book[] books) {

        Arrays.sort(books, Comparator.comparing(book -> book.title));
    }
}
```

```
// Binary Search
public static Book binarySearch(Book[] books, String title) {

    int low = 0;
    int high = books.length - 1;

    while (low <= high) {

        int mid = (low + high) / 2;

        int compare = books[mid].title.compareToIgnoreCase(title);

        if (compare == 0) {
            return books[mid];
        }
        else if (compare < 0) {
            low = mid + 1;
        }
        else {
            high = mid - 1;
        }
    }

    return null;
}
```

```
package ques6LibraryManagementSystem;
public class LibraryTest {

    public static void main(String[] args) {

        Book[] books = {

            new Book(101, "Java", "James Gosling"),
            new Book(102, "Python", "Guido van Rossum"),
            new Book(103, "C Programming", "Dennis Ritchie"),
            new Book(104, "Data Structures", "Mark Allen")
        };

        System.out.println("Linear Search:");

        Book book1 = LibraryManager.linearSearch(books, "Python");

        if (book1 != null)
            System.out.println(book1);

        LibraryManager.sortBooks(books);

        System.out.println("\nBinary Search:");

        Book book2 = LibraryManager.binarySearch(books, "Python");

        if (book2 != null)
            System.out.println(book2);
    }
}
```

Linear Search:

Book ID: 102, Title: Python, Author: Guido van Rossum

Binary Search:

Book ID: 102, Title: Python, Author: Guido van Rossum